

Academic programs

B. Tech - Civil Engineering

Program Educational Objectives (PEOs)

To provide students with a solid foundation in Basic and Engineering Sciences to understand, analyse and evaluate the information to achieve expertise in core areas of Civil Engineering.

To prepare the students to achieve high level technical expertise in the fields of Civil Engineering and to excel in the design and construction of various components or systems of Civil Engineering and to make the students capable of pursuing higher studies and research.

To establish acquaintance with the practical implementation of the theoretical concepts through laboratories, by bringing the real world into the academics through virtual industry labs and to enhance experimental skills of students beyond curriculum and encouraging them to identify and solve complex engineering problems.

To equip students with modern professional abilities such as effective communication, collaborative work in diverse teams, ethical decision making, successful management of personal and professional career objectives and passion for continuous development through lifelong learning.

To prepare the students to guide their professional development by bringing awareness of professional society activities, professional licensure requirements and opportunities for further education in graduate school.

Program Outcomes (POs)

Engineering Graduates will be able to:

Engineering Knowledge: The student is capable of applying the principles of basic sciences and mathematics in learning the Civil Engineering subjects. Graduates will be proficient in the core principles of Civil Engineering as they pertain to the sub-fields of Environmental Engineering, Geo-Technical Engineering, Structural Engineering, Transportation Engineering, Water Resources Engineering and will be able to apply these principles in engineering practice.

Problem Analysis: The Graduates will possess critical thinking skills, problem solving abilities and familiarity with the computational procedures essential to the field.

Design & Development of Solutions: The

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student is able to plan, analyze, design and look after the construction of various types of Civil Engineering structures with appropriate consideration for public health and safety, cultural, societal and environmental considerations.

Conduct investigations of complex problems:

The student will use research based methods to design, conduct experiments, to analyze and interpret experimental data.

Modern Tool Usage: The student will get

hands on training in the various modern Civil Engineering software and modern equipment.

The Engineer and Society: The students will

apply reasoning and uses appropriate knowledge to assess societal, health, safety, legal and cultural issues.

Environment and Sustainability: As the

students possess substantial knowledge in multi disciplinary subjects, they will be able to plan various projects keeping in view of its environmental effects on other related fields.

Ethics: The student will apply ethical

principles and commitment to profession and responsibilities of their profession.

Individual and Team work: The graduate is

capable of working productively as an individual, as a member or a leader in driver set teams.

Communication: The student will possess

mastery in good communication skills, expressing ideas, writing technical reports and effective managerial skills.

Project Management and Finance: The

graduates will acquire knowledge and understanding of the critical issues for professional practices such as the procurement of works, interaction with contractors during the construction phase of a project, financial and managerial capabilities.

Life-Long learning: The student will have

an awareness of contemporary issues and will contribute to the well being of the community with life-long learning in the broadest context of ever growing technology.

Program Specific Outcomes

Survey, Plot & Plan layout for Civil Engineering Structures and alignment for Canals & Roads

Analyze the problems related to structural components for

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Buildings, Pavements and Waterways and recommend suitable measures with appropriate consideration for public health, safety and Environmental sustainability.

Specify, design, supervise, test & evaluate foundations & superstructures for Buildings, Industries. Hydraulic structures, Powerhouses, Highways, Railways, Airways, Water supply systems & Sewage treatment plants.

M.Tech- Structural Engineering

To provide proficiency in the basic principles and advanced courses of technology in Structural Engineering so that students are able to formulate, analyse and solve the societal problems for sustainable development related to structural Engineering.

To expose the students to the latest innovations and trends with a view to inculcate strong research orientation in structural engineering as well as in multidisciplinary streams.

To produce Structural Engineers who integrate and build on the program's core curricular concepts in the pursuit of professional leadership, teamwork, life-long learning, and successful career advancement.

PO-1: An ability to independently carryout research/investigation and development work to solve practical problems.

PO-2: An ability to write and present a substantial technical report/document.

PO-3: Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

PO-4: To motivate the graduate students to address the societal needs by interdisciplinary approach through advanced courses.

PO-5: To enrich the graduate students to get hands on training on latest equipment / software to be industry ready / pursue advanced research.

PO-6: To inculcate ethical practices and to establish understanding of professionalism, safety, sustainability, their duties, and contribution to the society.

Program Specific Outcomes (PSOs)

Students will be able to:

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- To expose the graduate students to advanced courses in Analysis, and Design of RC, PSC and Steel structures as per the latest design codes, current national and international scenario on Structural Engineering.
- To motivate the graduate students to address the societal needs by interdisciplinary approach through advanced courses such as Finite Element Analysis, Plates & Shell structures, Structural Dynamics, Soil Dynamics, and allied courses.
- To enrich the graduate students to get hands on training on latest equipment / software to be industry ready / pursue advanced research.

M.Tech- Highway Engineering

- To provide the student with a solid foundation in Planning, Design, Construction, Operation and Maintenance of highway projects in the context of environmental, economic and social requirements.
- To equip students with modern concepts and tools which would enable them to become globally competent professionals and fitting into a broad range of career opportunities in highway research and industry.
- To prepare the students in lifelong learning for professional advancement, that would help them to execute complex transportation projects with professional ethics and social responsibilities.
- To expose graduate students to advanced courses in Pavement Material Characterization, Pavement Design, Traffic Planning, and Safety Engineering, Highway Construction Practices that comply with IRC design guidelines, as well as international Highway Engineering scenario.
- To encourage graduate students to address societal needs through advanced courses such as Computational Techniques in Highway Engineering, Highway Network Analysis and Optimisation Techniques, Artificial Intelligence and Big Data Analytics in Highway Engineering, and allied courses.
- To provide graduate students with hands-on training on cutting-edge equipment and software in order for them to be industry ready and pursue advanced research.

M.Tech - Geotechnical Engineering

- To apply knowledge of geotechnical engineering to produce engineers to integrate and build concepts to improve professional leadership, teamwork, life-long learning, and career advancement.
- To design and conduct experiments, to analyze and interpret

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data related to the geotechnical engineering, as well as to formulate systems within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

To accentuate the understanding of the basic principles and exposes the student to the latest developments, with a strong research foundation so as to provide engineering solutions in a global, economic, environmental, and societal context.

Identify, Investigate and Explore soil strata for construction of Buildings, Bridges, Dams, Retaining Walls, Pavements and Underground structures.

Analyze and Address the problems of foundations, slopes and retaining structures for different failure criteria, deformations and settlement behavior under static and dynamic loading pattern.

Design the substructures appropriately to overcome problematic soil behaviors such as collapsible, swelling and to support enormous structural loads.